

Appendix C - Geographe Marine Park Data Analysis Report

South-west Marine Parks Network

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Data Analysis

Statistical methods and model selection results underpinning the natural values reported in Appendix A.

Benthic ecosystem extent

Table C 1.1: Habitat model selection results. Top generalised additive models (GAMs) for predicting the probability of occurrence of key habitats from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the most parsimonious model (fewest variables - shown in bold) within two units of the lowest AICc.

Response	Model	ΔAICc	ωAICc	R^2	EDF
Macroalgae	detrended+aspect+Z+year	0	0.987	0.13012	12.10
Sand	detrended+aspect+Z+year	0	0.956	0.42942	10.94
Seagrass	detrended+aspect+Z+year	0	0.731	0.43034	11.64
Rock	detrended+roughness+as- pect+year	0	0.616	0.65959	10.86
	detrended+roughness+as- pect+year	1.741	0.258	0.66133	11.96
Sessile invertebrates	roughness+aspect+Z+year	0	0.465	0.17726	8.16
	roughness+aspect+Z+year	1.506	0.219	0.17713	8.94
Reef	detrended+roughness+as- pect+year	0	0.978	0.11303	12.93

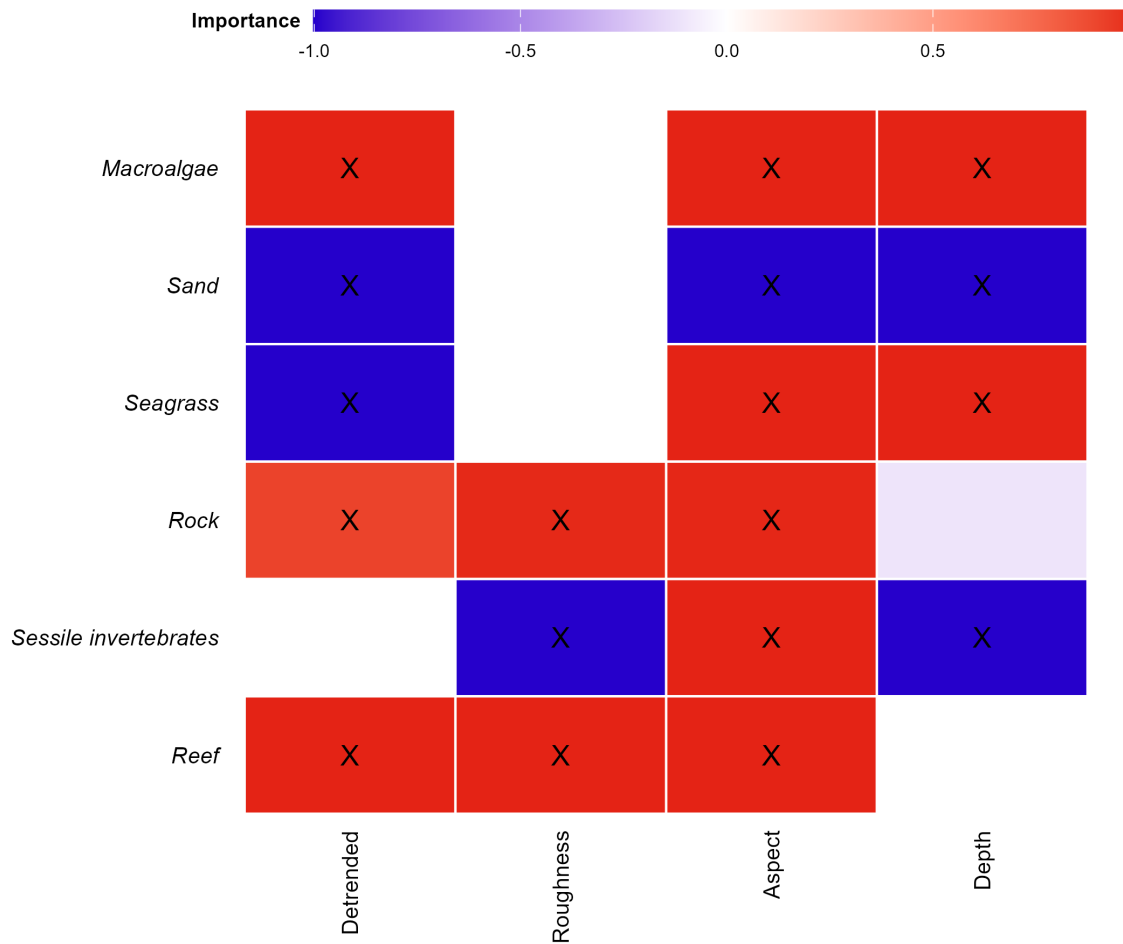


Figure C 1.1: Variable importance scores for predicting the probability of occurrence of main habitat categories. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient.

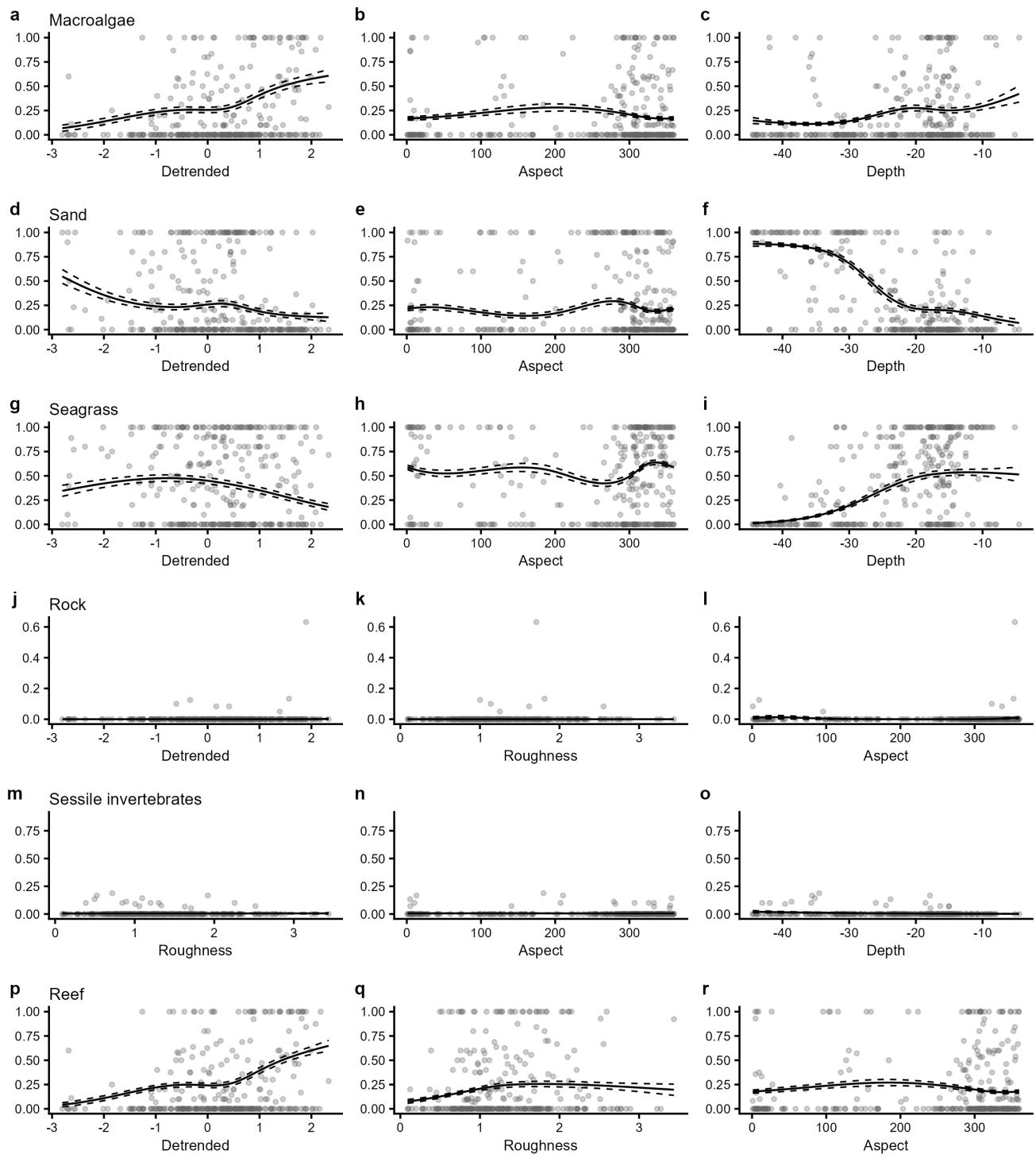


Figure C 1.2.1: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types in 2014. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data. Axis limits are held constant across 2014 and 2024 to allow direct comparison.

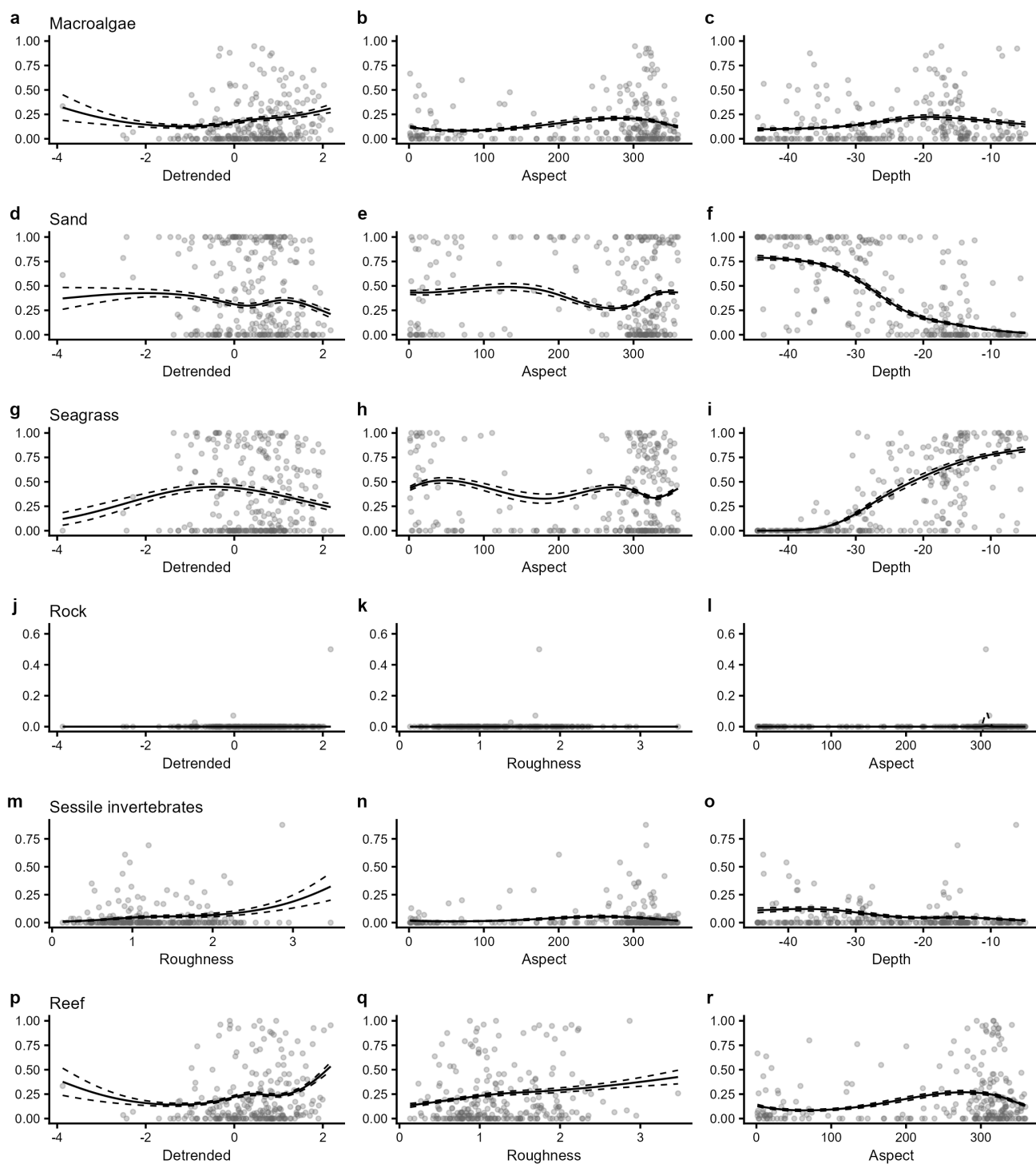


Figure C 1.2.2: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types in 2024. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data. Axis limits are held constant across 2014 and 2024 to allow direct comparison.

Demersal fish assemblage

Table C 2.1: Demersal fish model selection results. Top generalised additive models (GAMs) for predicting the species and abundance distribution of metrics of interest from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the best model (fewest variables - shown in bold) within two units of the lowest AICc. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Response	Model	ΔAICc	ωAICc	AICc	R^2	EDF
Species richness	roughness+reef+year+status	0	0.194	6513.26	0.16279	5.74
	roughness+aspect+reef+year+status	0	0.194	6513.26	0.16279	6.74
	reef+year+status	0.074	0.187	6513.34	0.15949	4.74
	aspect+reef+year+status	0.074	0.187	6513.34	0.15949	5.74
	detrended+roughness+reef+year+status	1.629	0.086	6514.89	0.16468	7.12
	detrended+reef+year+status	1.855	0.077	6515.12	0.16017	5.82
	detrended+aspect+reef+year+status	1.855	0.077	6515.12	0.16017	6.82
Total abundance	roughness+reef+year+status	0	0.194	6513.26	0.16279	5.74
	roughness+aspect+reef+year+status	0	0.194	6513.26	0.16279	6.74
	reef+year+status	0.074	0.187	6513.34	0.15949	4.74
	aspect+reef+year+status	0.074	0.187	6513.34	0.15949	5.74
	detrended+roughness+reef+year+status	1.629	0.086	6514.89	0.16468	7.12
	detrended+reef+year+status	1.855	0.077	6515.12	0.16017	5.82
	detrended+aspect+reef+year+status	1.855	0.077	6515.12	0.16017	6.82
Large Reef Fish Index*	detrended+aspect+Z+reef+year	0	0.397	7102.36	0.20843	7.97
	detrended+Z+reef+year	0.726	0.276	7103.09	0.20233	6.79
Reef Fish Thermal Index	detrended+aspect+reef+year+status	0	0.505	3188.42	0.3795	7.30
	detrended+aspect+Z+reef+year	1.643	0.222	3190.06	0.37756	7.57

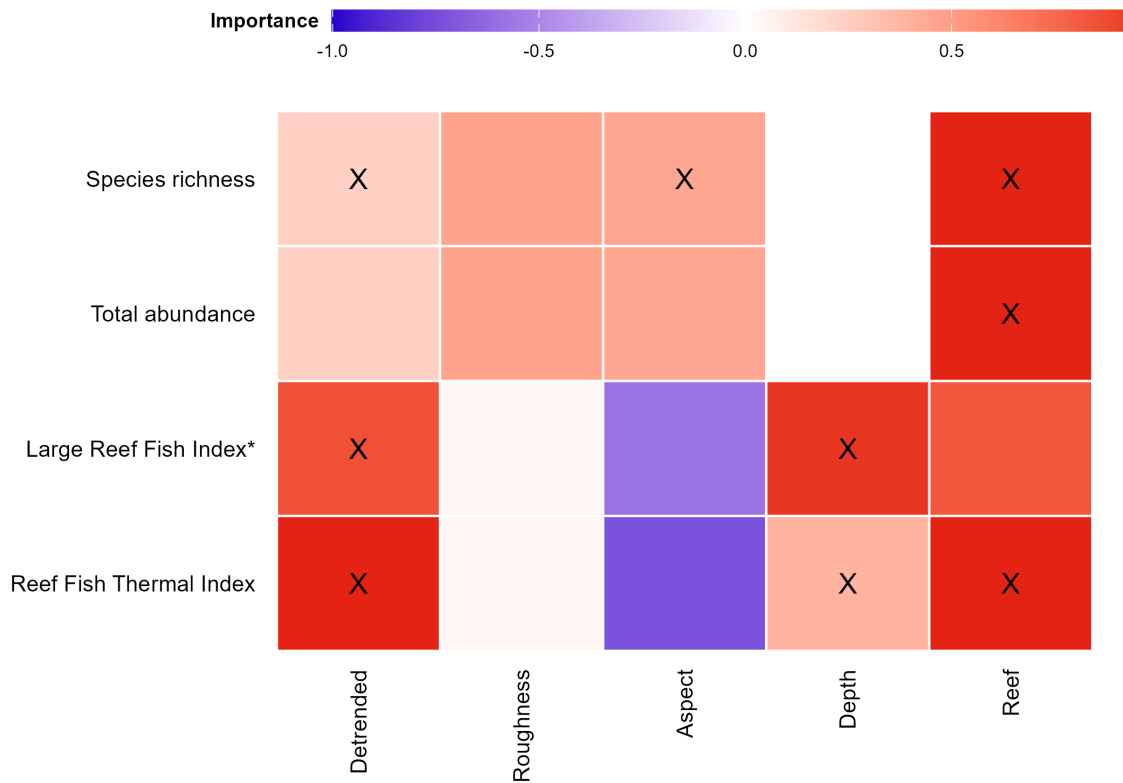


Figure C 2.1: Variable importance scores for predicting the probability of demersal fish metrics. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

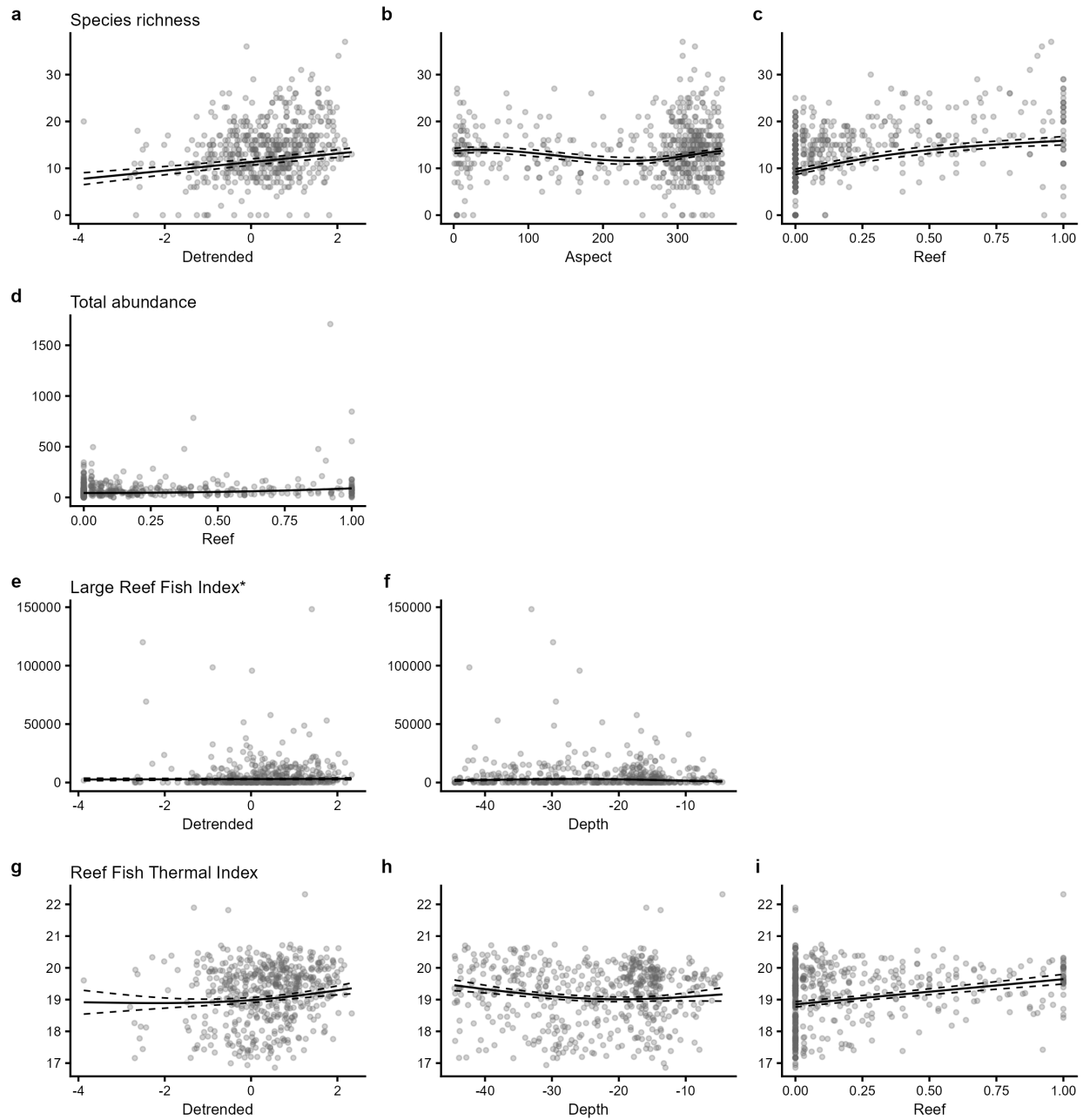


Figure C 2.2: Plots of the best model found to predict the probability of occurrence of key fish groups. Individual panes display the predicted relationships between model covariates and each demersal fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value, with year held at 2014 and zone status held at areas open to fishing. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years and zones. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.